

This project investigates a novel soft robotic exoskeleton design to enhance hand function in individuals dealing with hand or finger muscle weakness. With a combined soft-fabric pneumatic actuator and a bio-inspired granular-and-laminar jamming module, the soft exoskeleton aims to achieve a biomimetic range of motion by varying the pneumatic actuator's air pressure across 8 distinct jamming configurations, with a similar or greater degree of stiffness modulation compared to other designs. The pneumatic flexion will be achieved with an impulse-sealed air pocket inside a pleated fabric pocket with stiff fabric lining the bottom layer, and elastic fabric forming the top pleats. The jamming module will be silicone-molded following an H-block design, in which rounded granular pockets serve as joints, and long laminar layers act as bones slotted into the pockets to optimize load distribution along the exoskeleton's length. The study will assess the feasibility of this design as a wearable exoskeleton, including overall device weight, ROM, actuation speed, stiffness enhancement, and maximum force output, with and without the jamming module. The aim is to contribute data on the viability and limits of jamming mechanisms for enhancing soft robotic exoskeletons, as well as to provide insight into different control methods that could optimize cost, computation speed, and functionality for soft robotic exoskeletons.